

Review Exam I

Complex Analysis

Underlined Definitions: May be asked for on exam

Underlined Propositions or Theorems: Proofs may be asked for on exam

Cauchy's Integral Formula (#0). Let G be a region in $\mathbb{C}$, let $\overline{B(a,r)} \subset G$ and let g be the circle $C(a,r)$, oriented positively. Let $f \in A(G)$. Then, for $z \in B(a,r)$, $f(z) = \frac{1}{2\pi i} \int_g \frac{f(w)}{w-z} dw$.

Cauchy's Theorem (#0). Let G be a region in $\mathbb{C}$, let $B(a,r) \subset G$ and let g be a closed rectifiable curve in $B(a,r)$. Let $f \in A(G)$. Then, $\int_g f = 0$.

Chapter 4.5

Definition. $g \approx 0$ for a closed rectifiable curve g lying in a region G if ...

Lemma 5.1. Let g be a rectifiable curve and let j be continuous on $\{g\}$. For each $m \geq 1$, let

$$f_m(z) = \int_g \frac{j(w)}{(w-z)^m} dw \text{ for } z \in \mathbb{C} \setminus \{g\}. \text{ Then, } f_m \in A(\mathbb{C} \setminus \{g\}) \text{ and } f'_m(z) = m f_{m+1}(z).$$

Cauchy's Integral Formula (#1). Let G be a region in $\mathbb{C}$ and let $f \in A(G)$. Let g be a closed rectifiable curve in G such that $g \approx 0$. Then, for $z \in G \setminus \{g\}$, $n(g; z) f(z) = \frac{1}{2\pi i} \int_g \frac{f(w)}{w-z} dw$.

Cauchy's Integral Formula (#2). Let G be a region in $\mathbb{C}$ and let $f \in A(G)$. Let $g_1, g_2, \dots, g_m$ be a closed rectifiable curves in G such that $g_1 + g_2 + \dots + g_m \approx 0$. Then, for $z \in G \setminus \bigcup_{k=1}^m \{g_k\}$,

$$\sum_{k=1}^m n(g_k; z) f(z) = \sum_{k=1}^m \frac{1}{2\pi i} \int_{g_k} \frac{f(w)}{w-z} dw.$$

Cauchy's Theorem (#1). Let G be a region in $\mathbb{C}$ and let $f \in A(G)$. Let $g_1, g_2, \dots, g_m$ be a closed rectifiable curves in G such that $g_1 + g_2 + \dots + g_m \approx 0$. Then, $\sum_{k=1}^m \int_{g_k} f = 0$

Corollary. Let G be a region in $\mathbb{C}$ and let $f \in A(G)$. Let g be a closed rectifiable curve in G such that $g \approx 0$.

$$\text{Then, for } z \in G \setminus \{g\}, n(g; z) f^{(k)}(z) = \frac{k!}{2\pi i} \int_g \frac{f(w)}{(w-z)^{k+1}} dw.$$

Morera's Theorem Let G be a region in $\mathbb{C}$ and let $f \in C(G)$. Suppose that $\int_T f = 0$ for every triangular path $T \subset G$. Then, $f \in A(G)$.

Chapter 4.6

Definition. $g_0 \sim g_1$ for two closed rectifiable paths g_0, g_1 lying in a region G if ...

Proposition. “ $\sim$ ” is an equivalence relation

Examples of pairs of curves g_0, g_1 (and the overlying region G) which are homotopic and of pairs of curves g_0, g_1 (and the overlying region G) which are not homotopic

Problems about constructing the homotopy function for a pair of curves g_0, g_1 (and the overlying region G) which are homotopic

Definition. A set S is convex if ...

Definition. A set S is starlike (w.r.t. a) if ...

Examples of sets which are convex / starlike and of sets which are not convex / starlike

Cauchy's Theorem (#2). Let G be a region in $\mathbb{C}$ and let g be a closed rectifiable curve in G . If $g \sim 0$, then $\int_g f = 0$.

Corollary. Let G be a region in $\mathbb{C}$ which is starlike and let $f \in A(G)$. Let g be a closed rectifiable curve in G . Then, $\int_g f = 0$.

Cauchy's Theorem (#3). Let G be a region in $\mathbb{C}$ and let $f \in A(G)$. Let g_0, g_1 be closed rectifiable curves in G such that $g_0 \sim g_1$. Then, $\int_{g_0} f = \int_{g_1} f$.

Corollary. Let G be a region in $\mathbb{C}$ and let $f \in A(G)$. Let g be a closed rectifiable curve in G such that $g \sim 0$. Then, $\int_g f = 0$.

Definition. $g_0 \sim_{FEP} g_1$ for two rectifiable paths g_0, g_1 lying in a region G if ...

Examples of pairs of curves g_0, g_1 (and the overlying region G) which are FEP-homotopic and of pairs of curves g_0, g_1 (and the overlying region G) which are not FEP-homotopic

Independence of Path Theorem. Let G be a region in $\mathbb{C}$ and let $f \in A(G)$. Let g_0, g_1 be rectifiable curves in G such that $g_0 \sim_{FEP} g_1$. Then, $\int_{g_0} f = \int_{g_1} f$.

Definition. A region G is simply connected if ...

Examples of regions which are simply connected and of regions which are not simply connected

Cauchy's Theorem (#4). Let G be a region in $\mathbb{C}$ which is simply connected and let $f \in A(G)$. Let g be a closed rectifiable curve in G . Then, $\int_g f = 0$.

Corollary. Let G be a region in $\mathbb{C}$ which is simply connected and let $f \in A(G)$. Then, f has a primitive on G .

Definition. A function f has a branch-of-log f on a region G if ...

Corollary. Let G be a region in $\mathbb{C}$ which is simply connected and let $f \in A(G)$. If $f(z) \neq 0$ on G , then f has a branch-of-log f on G .

Chapter 4.7

Theorem 7.2. Let G be a region in $\mathbb{C}$ and let $f \in A(G)$. Suppose $Z_f = \{a_1, a_2, \dots, a_m\}$. Let g be a closed rectifiable curve in G such that $g \approx 0$ and such that $Z_f \cap \{g\} = \emptyset$. Then $\frac{1}{2\pi i} \int_g \frac{f'(z)}{f(z)} dz = \sum_{k=1}^m n(g; a_k)$

Corollary 7.3. Let G be a region in $\mathbb{C}$ and let $f \in A(G)$. Suppose $Z_{f-a} = \{a_1, a_2, \dots, a_m\}$. Let g be a closed rectifiable curve in G such that $g \approx 0$ and such that $Z_{f-a} \cap \{g\} = \emptyset$. Then $\frac{1}{2\pi i} \int_g \frac{f'(z)}{f(z)-a} dz = \sum_{k=1}^m n(g; a_k)$

Proposition. Let G be a region in $\mathbb{C}$ and let $f \in A(G)$. Let g be a closed rectifiable curve in G such that $g \approx 0$. Let $s = f \circ g$. Let a, b belong to the same component of $\mathbb{C} \setminus \{s\}$. Then,

$$\begin{aligned} n(s; a) &= n(s; b) \\ \parallel &\parallel \\ \sum_{Z_{f-a}} n(g; z_k(a)) &= \sum_{Z_{f-b}} n(g; z_k(b)) \end{aligned}$$

Definition. A function f has a simple zero at $z = a$ if ...

Theorem 7.4. Let G be a region in $\mathbb{C}$, let $B(a, r) \subset G$ and let $f \in A(G)$. Let $f(a) = a$. If $f(z) - a$ has a zero of order m at $z = a$, then there exists $\epsilon > 0$ and $d > 0$ such that for each $z \in B(a, d)$ the equation $f(z) = z$ has exactly m simple roots in $B(a, \epsilon)$.

Corollary. Open Mapping Theorem

Chapter 4.8

Cauchy-Goursat Theorem. Let G be a region. If $f \in D(G)$, then $f \in A(G)$.

Chapter 5.1

Definition. A function f has an isolated singularity at $z = a$ if . . .

Examples of functions which have isolated singularities

Definition. Removable Singularity, Pole, Essential Singularity

Examples of functions which have removable singularities, poles, essential singularities

Problems about classifying singularities as removable singularities, poles, essential singularities

Proposition. Let G be a region in $\mathbb{C}$ and let f have an isolated singularity at $z = a$, $a \in G$. The following are equivalent.

- a. f has a removable singularity at $z = a$
- b. f is bounded (in modulus) on a punctured neighborhood of a
- c. $\lim_{z \rightarrow a} f(z)$ exists
- d. $\lim_{z \rightarrow a} (z - a) f(z) = 0$

Definition. A function f has a pole of order m at $z = a$ if . . .

Proposition. Let G be a region in $\mathbb{C}$ and let f have an isolated singularity at $z = a$. f has a pole of order m at $z = a$ if and only if there exists a function g analytic and non-vanishing on a neighborhood of $z = a$ such that

$$f(z) = \frac{g(z)}{(z - a)^m}.$$

Definition. The singular part of a function f which has a pole of order m at $z = a$ is . . .

Problems about identifying the singular parts of functions with poles

Definition. A double infinite series $\sum_{n=-\infty}^{\infty} z_n$ is absolutely convergent if . . .

Definition. A double infinite series $\sum_{n=-\infty}^{\infty} u_n(s)$ is uniformly convergent on a region S if . . .

Examples of double infinite series which are absolutely convergent / uniformly convergent.

Definition. $ann(a; r_1, r_2) =$

Theorem (Laurent Series Expansion). Let $f \in A(\text{ann}(a; R_1, R_2))$. Let $G = \mathbb{C} \setminus \overline{B(a, R_1)}$. Then there exist functions $f_2 \in A(B(a, R_2))$ and $f_1 \in A(G)$ such that $f = f_2 + f_1$ on $\text{ann}(a; R_1, R_2)$. Furthermore,

$$f_2(z) = \sum_{n=0}^{\infty} a_n (z-a)^n \text{ on } B(a, R_2) \quad (2)$$

and

$$f_1(z) = \sum_{n=1}^{\infty} a_{-n} (z-a)^{-n} \text{ on } G \quad (1)$$

where

$$a_n = \frac{1}{2\pi i} \int_{C(a, r)} \frac{f(w)}{(w-a)^{n+1}} dw, \quad R_1 < r < R_2 \quad (*)$$

and the coefficient a_n in (*) is independent of the choice of r . The series for f_2 in (2) converges absolutely on $B(a, R_2)$ and uniformly on subsets $\overline{B(a, r_2)}$ of $B(a, R_2)$, $0 < r_2 < R_2$. The series for f_1 in (1) converges absolutely on G and uniformly on subsets $\overline{\text{ann}(a; r_1, \infty)}$ of G .

Furthermore, the series representation $f(z) = \sum_{n=-\infty}^{\infty} a_n (z-a)^n$ on $\text{ann}(a; R_1, R_2)$ is unique.

Corollary. Let G be a region in $\mathbb{C}$ and let f have an isolated singularity at $z = a$. Let $f(z) = \sum_{n=-\infty}^{\infty} a_n (z-a)^n$ be the Laurent Series expansion of f on a punctured neighborhood of a . Then,

- $z = a$ is removable singularity of f in and only if $a_n = 0$ for all $n \leq -1$
- $z = a$ is a pole of f of order m if and only if $a_{-m} \neq 0$ and $a_{-n} = 0$ for all $n \leq -m-1$
- $z = a$ is an essential singularity if and only if $a_n \neq 0$ for infinitely many negative n

Problems constructing Laurent Series expansions.

Problems constructing Laurent Series expansions for rational functions.

Casorati-Weierstrass Theorem. Let G be a region in $\mathbb{C}$ and let f have an isolated singularity at $z = a$. If $z = a$ is an essential singularity of f , then for every $d > 0$, $f(B(a, d)') = \mathbb{C}$.